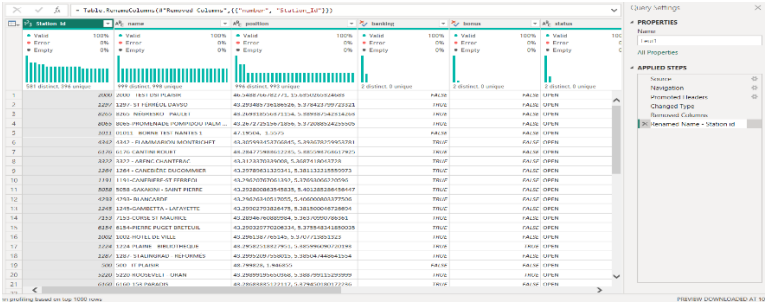
Smart Bike Sharing System Performance & Utilization Analysis

Cleaning and Transformation

Power Query

Renamed the id to station Id



Cleaned Latitude and Longitude

1.2 Latitude	1.2 Longitude
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
995 distinct, 991 unique	996 distinct, 993 unique
46.54887668	15.63502653
43.29348574	5.3784238
43.26931856	5.389387542
43.26727252	5.372088524
47.19504	-1.5575
43.30599345	5.39367826
43.28477598	5.385593764
43.31233703	5.368741804
43.29789631	5.381132216
43.29620767	5.376930662
43.29280086	5.401285286
43.29626341	5.406000803
43.29902794	5.381500047
43.28946761	5.363709908
43.29032977	5.375548342
43.29613878	5.370771385
43.29582518	5.385996091
43.29952098	5.385047449
48.799828	1.946855
43.29899196	5.388799115
43.28683835	5.37945018

Cleaned City

City
Valid 100%
Error 0%
Empty 0%
25 distinct, 0 unique
Maribor
Marseille
Marseille
Marseille
Nantes
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Marseille
Jcdecauxbike

Changed Banking to Payment Enabled

Payment_Enables
Valid 100%
Error 0%
Empty 0%
2 distinct, 0 unique
No
Yes
Yes
Yes
No
Yes
Yes
Yes
Yes
Yes
Yes
Yes
Yes
Yes
Yes
Yes
Yes
No

Dim_station

Queries [2]

- Bike Sharing
- Dim_Station

```
= Table.Distinct(#"Removed Columns", {"Station_Id"})
```

	Station_Id	Station_Name	Latitude	Longitude	Payment_Enables	Bonus_Status	City
Valid	100%	100%	100%	100%	100%	100%	100%
Error	0%	0%	0%	0%	0%	0%	0%
Empty	0%	0%	0%	0%	0%	0%	0%
	1000 distinct, 1000 unique	836 distinct, 833 unique	996 distinct, 993 unique	998 distinct, 996 unique	2 distinct, 0 unique	2 distinct, 0 unique	25 distinct, E
1	2000	Test Dsi Plaisir	46.54887668	15.63502653	No	No	Maribor
2	1297	St Férréol Davso	43.29348574	5.3784238	Yes	No	Marseille
3	8265	Négresko - Paulet	43.26931856	5.389387542	Yes	No	Marseille
4	8065	Promenade Pompidou Palm Beach	43.26727252	5.372088524	Yes	No	Marseille
5	1011	Borne Test Nantes 1	47.19504	-1.5575	No	No	Nantes
6	4342	Flammarion Montrichet	43.30599345	5.39367826	Yes	No	Marseille
7	6176	Cantini Rouet	43.28477598	5.385593764	Yes	No	Marseille
8	3322	Arenc Chanterac	43.31233703	5.368741804	Yes	No	Marseille
9	1264	Canebière Dugommier	43.29789631	5.381132216	Yes	No	Marseille
10	1191	Canebiere-St Ferreol	43.29620767	5.376930662	Yes	No	Marseille
11	5058	Sakakini - Saint Pierre	43.29280086	5.401285286	Yes	No	Marseille
12	4293	Blancarde	43.29626341	5.406000803	Yes	No	Marseille
13	1245	Gambetta - Lafayette	43.29902794	5.381500047	Yes	No	Marseille
14	7153	Corse St Maurice	43.28946761	5.363709908	Yes	No	Marseille
15	6154	Pierre Puget Breteuil	43.29032977	5.375548342	Yes	No	Marseille
16	1002	Hotel De Ville	43.29613878	5.370771385	Yes	No	Marseille
17	1224	Plaine - Bibliotheque	43.29582518	5.385996091	Yes	Yes	Marseille
18	1287	Stalingrad - Réformés	43.29952098	5.385047449	Yes	No	Marseille
19	500	It Plaisir	48.799828	1.946855	No	No	Jcdecaux
20	5220	Roosevelt - Oran	43.29899196	5.388799115	Yes	Yes	Marseille
21	6160	153 Paradis	43.28683835	5.37945018	Yes	No	Marseille

Dim_Date

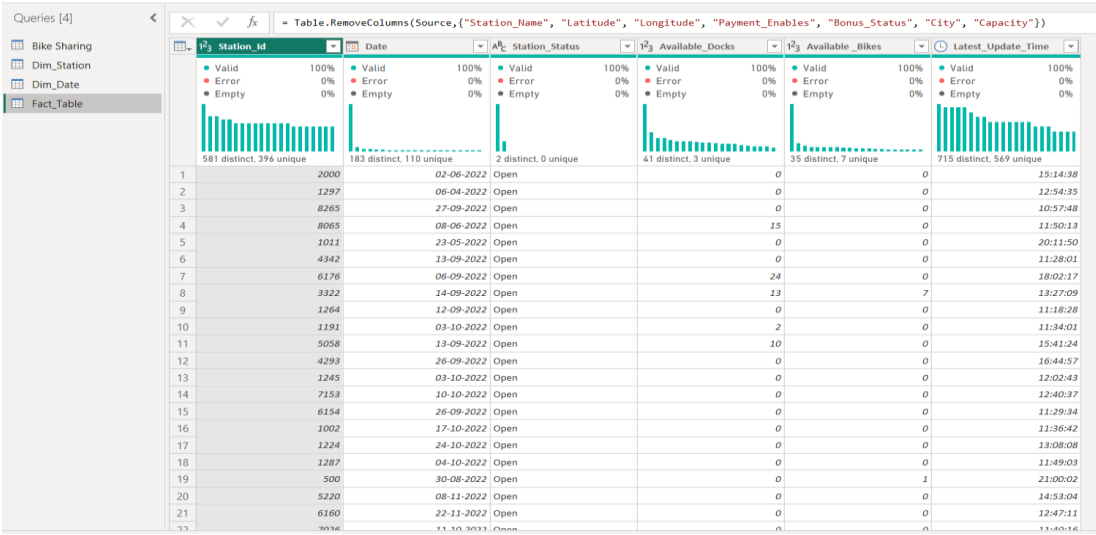
Queries [3]

- Bike Sharing
- Dim_Station
- Dim_Date

```
= Table.TransformColumnTypes(#"Removed Columns1",{{"Month",
```

	Date	Year	Month
Valid	100%	100%	100%
Error	0%	0%	0%
Empty	0%	0%	0%
	183 distinct, 183 unique	4 distinct, 0 unique	12 distinct, 0 unique
1	02-06-2022	2022	6
2	06-04-2022	2022	4
3	27-09-2022	2022	9
4	08-06-2022	2022	6
5	23-05-2022	2022	5
6	13-09-2022	2022	9
7	06-09-2022	2022	9
8	14-09-2022	2022	9
9	12-09-2022	2022	9
10	03-10-2022	2022	10
11	26-09-2022	2022	9
12	10-10-2022	2022	10
13	17-10-2022	2022	10
14	24-10-2022	2022	10
15	04-10-2022	2022	10
16	30-08-2022	2022	8
17	08-11-2022	2022	11
18	22-11-2022	2022	11
19	11-10-2022	2022	10
20	28-11-2022	2022	11
21	29-11-2022	2022	11

Fact_Table



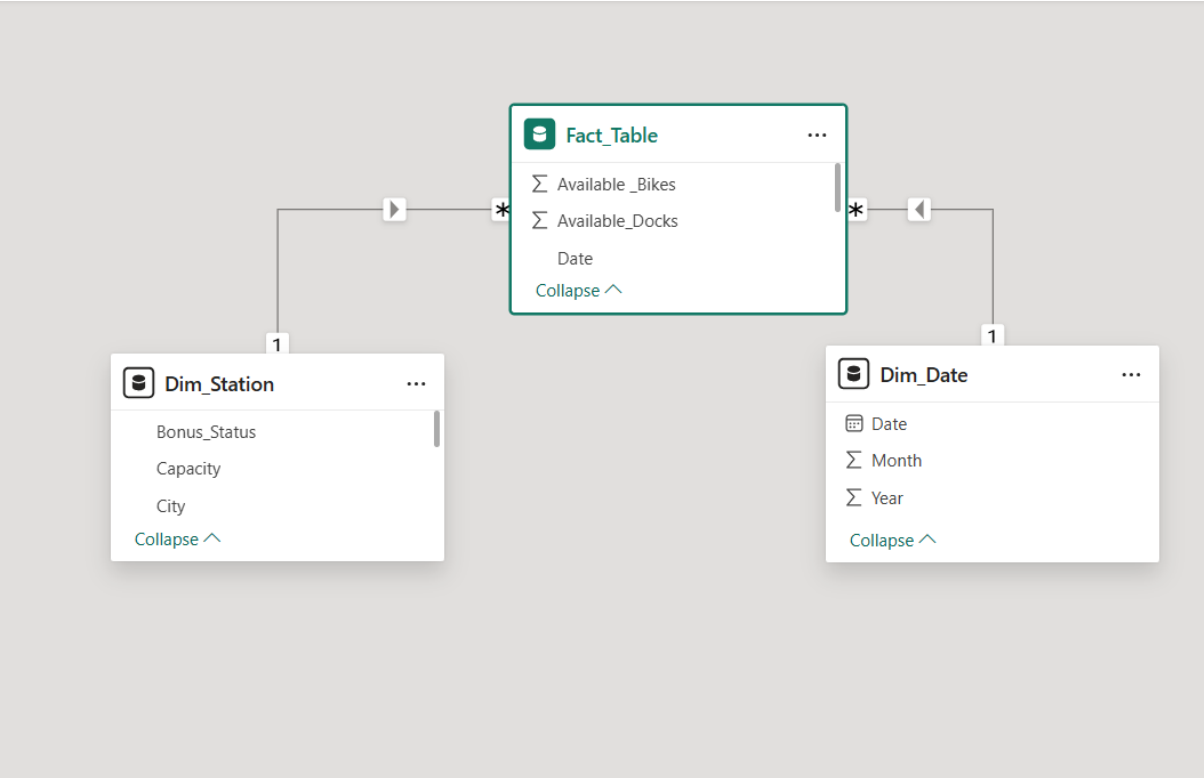
Queries [4]

- Bike Sharing
- Dim_Station
- Dim_Date
- Fact_Table

= Table.RemoveColumns(Source,{"Station_Name", "Latitude", "Longitude", "Payment_Enables", "Bonus_Status", "City", "Capacity"})

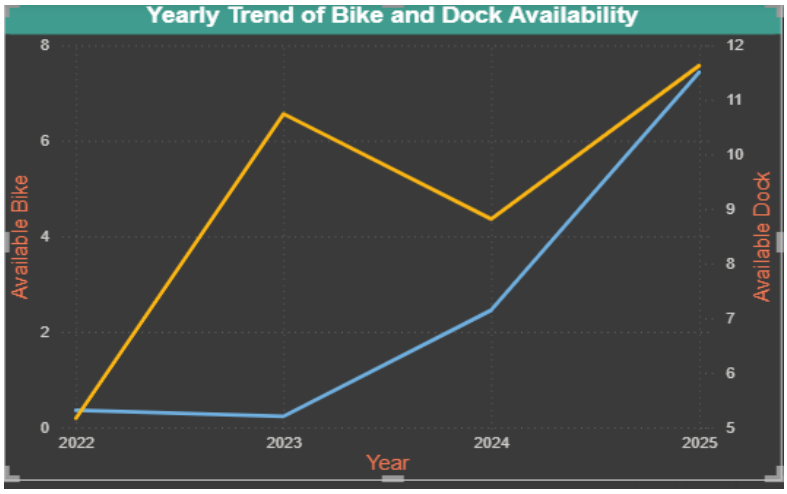
	Station_Id	Date	Station_Status	Available_Docks	Available _Bikes	Latest_Update_Time
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	581 distinct, 396 unique	183 distinct, 110 unique	2 distinct, 0 unique	41 distinct, 3 unique	35 distinct, 7 unique	715 distinct, 569 unique
1	2000	02-06-2022	Open	0	0	15:14:38
2	1297	06-04-2022	Open	0	0	12:54:35
3	8265	27-09-2022	Open	0	0	10:57:48
4	8065	08-06-2022	Open	15	0	11:50:13
5	1011	23-05-2022	Open	0	0	20:11:50
6	4342	13-09-2022	Open	0	0	11:28:01
7	6176	06-09-2022	Open	24	0	18:02:17
8	3322	14-09-2022	Open	13	7	13:27:09
9	1264	12-09-2022	Open	0	0	11:18:28
10	1191	03-10-2022	Open	2	0	11:34:01
11	5058	13-09-2022	Open	10	0	15:41:24
12	4293	26-09-2022	Open	0	0	16:44:57
13	1245	03-10-2022	Open	0	0	12:02:43
14	7153	10-10-2022	Open	0	0	12:40:37
15	6154	26-09-2022	Open	0	0	11:29:34
16	1002	17-10-2022	Open	0	0	11:36:42
17	1224	24-10-2022	Open	0	0	13:08:08
18	1287	04-10-2022	Open	0	0	11:49:03
19	500	30-08-2022	Open	0	1	21:00:02
20	5220	08-11-2022	Open	0	0	14:53:04
21	6160	22-11-2022	Open	0	0	12:47:11

Data Modeling



Visualization

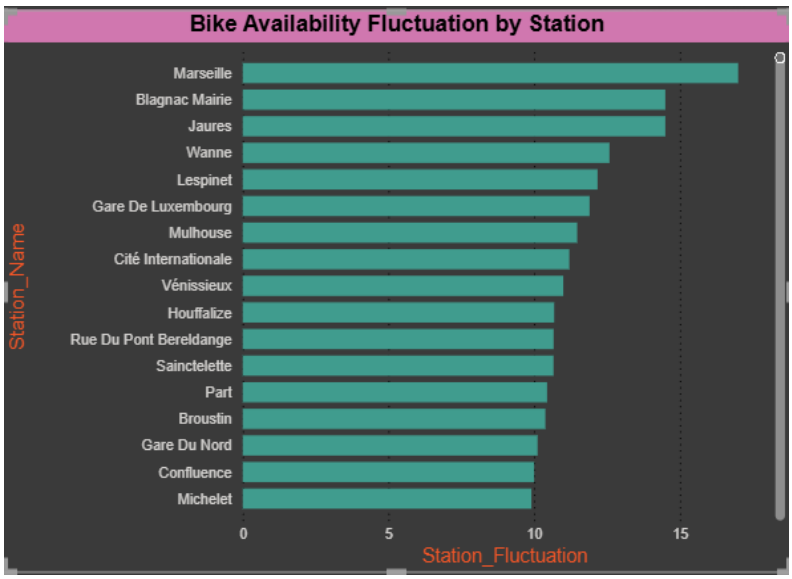
Yearly Trend of Bike and Dock Availability



Insights

- Average available bikes increased from ~0.3 in 2022 to ~7.8 in 2025 (partial), showing improved bike availability.
- Average available docks peaked at ~11.5 in 2025 (partial), indicating better return capacity compared to previous years.

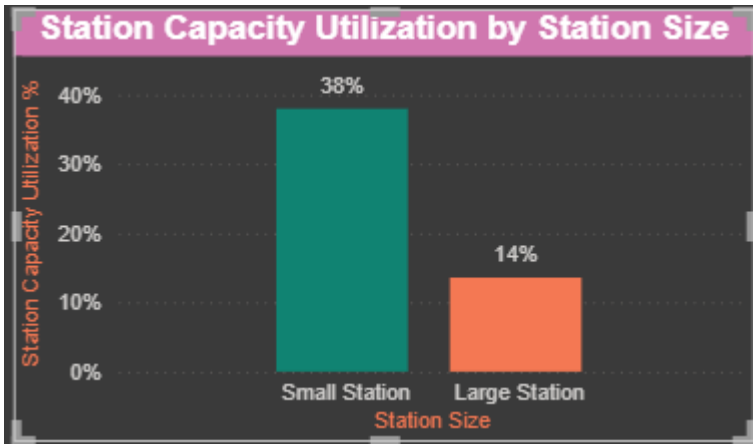
Bike Availability Fluctuation By Station



Insights

- Stations such as Marseille and Blagnac Mairie show the highest bike availability fluctuation (>15), indicating inconsistent supply-demand patterns and a need for frequent rebalancing.

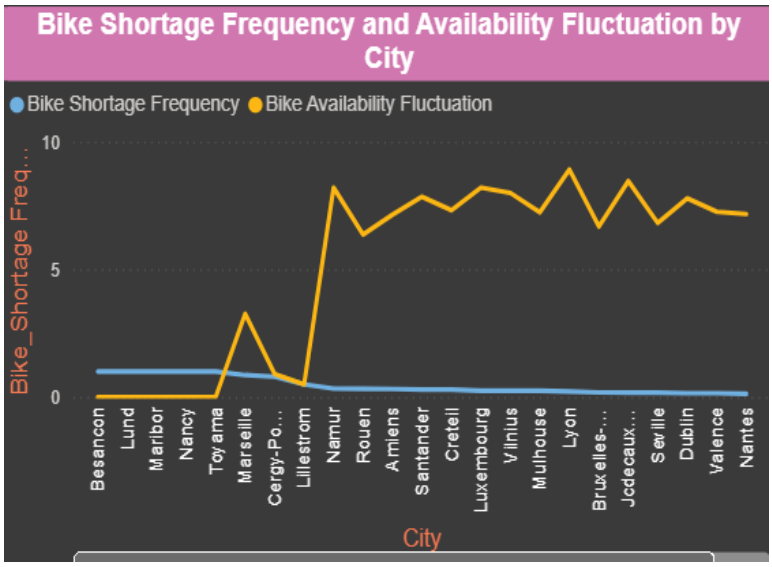
Station Capacity Utilization by Station Size



Insights

- Small stations operate at nearly 38% capacity utilization, almost three times higher than large stations (14%), indicating better efficiency.

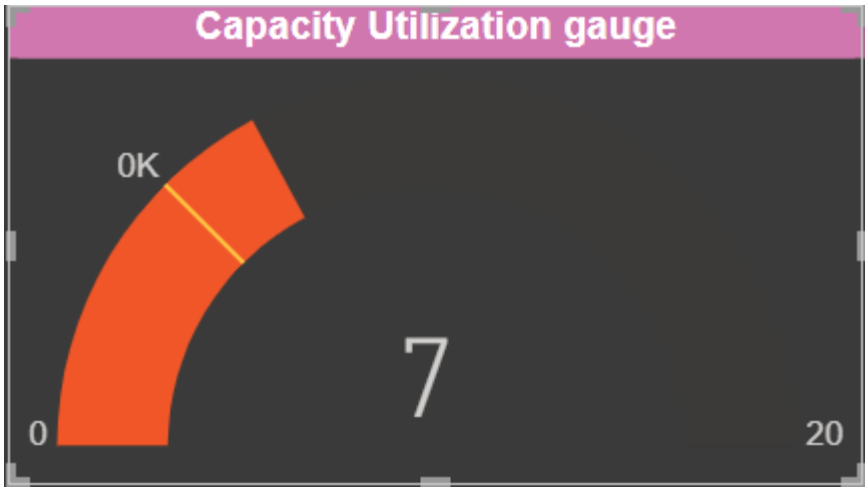
Bike Shortage Frequency and Availability Fluctuation by City



Insight

- Cities such as Brussels-Capitale and Lyon exhibit high bike availability fluctuation (≈8–9) along with recurring shortages, signaling the need for capacity expansion.

Capacity Utilization gauge



Insight

- The average bike availability is 7 bikes per station, exceeding the target of 5 and indicating healthy system performance.